MATH-517: Assignment 3

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Theoretical exercise

Local Linear Regression as a Linear Smoother

We are given i.i.d samples (x_i, y_i), $i = 1, \dots, n$ from the regression model

$$y_i = m(x_i) + \varepsilon_i, \quad i = 1, \dots, n$$

where the predictors are one-dimensional and $\mathbb{E}[\varepsilon_i] = 0$.
The goal is to estimate the regression function m and show that $\hat{m}$ is a linear smoother as seen in the course.

1.1 Weighted average representation

We show that $\hat{m}(x)$ can be expressed as a weighted average of the observations:

$$\hat{m}(x) = \sum_{i=1}^{n} w_{ni}(x) Y_i,$$

where the weights $w_{ni}(x)$ depend only on x, X_i, K and h, but not on the Y_i's.

We consider the weighted least squares problem

$$(\hat{\beta}_0(x), \hat{\beta}_1(x)) = \arg\min_{\beta_0, \beta_1} \sum_{i=1}^{n} \left(Y_i - \beta_0 - \beta_1(X_i - x)\right)^2 K\left(\frac{X_i - x}{h}\right).$$

Notations:

- Let Z be the $n \times 2$ design matrix with rows $z_i^T = (1, X_i - x)$.

- Let $W = \mathrm{diag}\Big(K\Big(\frac{X_1 - x}{h}\Big), \dots, K\Big(\frac{X_n - x}{h}\Big)\Big)$.
- Let $Y = (Y_1, \dots, Y_n)^T$.

Then the weighted least squares solution is

$$\hat{\beta}(x) = (\hat{\beta}_0(x), \hat{\beta}_1(x))^T = (Z^T W Z)^{-1} Z^T W Y.$$

The fitted value at x is

$$\hat{m}(x) = \hat{\beta}_0(x) = e_1^T \hat{\beta}(x) = e_1^T (Z^T W Z)^{-1} Z^T W Y,$$

where $e_1 = (1, 0)^T$.

Hence

$$\hat{m}(x) = e_1^T (Z^T W Z)^{-1} Z^T W Y = \sum_{i=1}^{n} \Big[e_1^T (Z^T W Z)^{-1} Z^T W \Big]_i Y_i,$$

so that

$$\hat{m}(x) = \sum_{i=1}^{n} w_{ni}(x) Y_i,$$

with weights

$$w_{ni}(x) = \Big[e_1^T (Z^T W Z)^{-1} Z^T W \Big]_i$$

which depend only on x, X_i, K and h as desired.

1.2 Explicit expression for the weights

We use the notation

$$S_{n,k}(x) = \frac{1}{nh} \sum_{i=1}^{n} (X_i - x)^k K\Big(\frac{X_i - x}{h}\Big), \quad k = 0, 1, 2$$

to derive an explicit expression for $w_{ni}(x)$ in terms of $S_{n,0}(x), S_{n,1}(x), S_{n,2}(x)$, and the kernel.

Set $k_i := K\Big(\frac{X_i - x}{h}\Big)$ and observe that

$$Z = \begin{pmatrix} 1 & X_1 - x \\ 1 & X_2 - x \\ \vdots & \vdots \\ 1 & X_n - x \end{pmatrix}, \qquad W = \operatorname{diag}(k_1, \dots, k_n).$$

$$Z^T W Z = \begin{pmatrix} 1 & \cdots & 1 \\ X_1 - x & \cdots & X_n - x \end{pmatrix} \begin{pmatrix} k_1 & & 0 \\ & \ddots & \\ 0 & & k_n \end{pmatrix} \begin{pmatrix} 1 & X_1 - x \\ \vdots & \vdots \\ 1 & X_n - x \end{pmatrix}.$$

$$= \begin{pmatrix} \sum_{i=1}^n k_i & \sum_{i=1}^n (X_i - x) k_i \\ \sum_{i=1}^n (X_i - x) k_i & \sum_{i=1}^n (X_i - x)^2 k_i \end{pmatrix}.$$

So if we denote $s_k := \sum_{i=1}^n (X_i - x)^k k_i = nh\, S_{n,k}(x), \qquad k = 0, 1, 2$, we obtain

$$Z^T W Z = \begin{pmatrix} s_0 & s_1 \\ s_1 & s_2 \end{pmatrix}, \qquad (Z^T W Z)^{-1} = \frac{1}{s_0 s_2 - s_1^2} \begin{pmatrix} s_2 & -s_1 \\ -s_1 & s_0 \end{pmatrix}.$$

Moreover,

$$Z^T W = \begin{pmatrix} 1 & \cdots & 1 \\ X_1 - x & \cdots & X_n - x \end{pmatrix} \begin{pmatrix} k_1 & & 0 \\ & \ddots & \\ 0 & & k_n \end{pmatrix} = \begin{pmatrix} k_1 & k_2 & \cdots & k_n \\ (X_1 - x) k_1 & (X_2 - x) k_2 & \cdots & (X_n - x) k_n \end{pmatrix}.$$

Therefore,

$$w_{ni}(x) = \left[(1, 0) \, \frac{1}{s_0 s_2 - s_1^2} \begin{pmatrix} s_2 & -s_1 \\ -s_1 & s_0 \end{pmatrix} \begin{pmatrix} k_1 & \cdots & k_n \\ (x_1 - x) k_1 & \cdots & (x_n - x) k_n \end{pmatrix} \right]_i$$

$$= \left[\frac{1}{s_0 s_2 - s_1^2} \begin{pmatrix} s_2 k_i - s_1 (X_1 - x) k_i & \cdots & s_n k_n - s_1 (X_n - x) k_n \end{pmatrix} \right]_i$$

The weight associated to observation i is therefore

$$w_{ni}(x) = \frac{1}{s_0 s_2 - s_1^2} \Big(s_2 K\Big(\tfrac{X_i - x}{h}\Big) - s_1 (X_i - x) K\Big(\tfrac{X_i - x}{h}\Big) \Big)$$

$$= \frac{nh\, S_{n,2}(x)\, K\Big(\tfrac{X_i - x}{h}\Big) - nh\, S_{n,1}(x) (X_i - x) K\Big(\tfrac{X_i - x}{h}\Big)}{nh\, S_{n,0}(x) nh\, S_{n,2}(x) - n^2 h^2 \, (S_{n,1}(x))^2}$$

$$= \frac{1}{nh} \, \frac{K\Big(\tfrac{X_i - x}{h}\Big) \, (S_{n,2}(x) - (X_i - x) S_{n,1}(x))}{S_{n,0}(x) S_{n,2}(x) - (S_{n,1}(x))^2}.$$

1.3 Normalization property

Summing over all i,

$$\sum_{i=1}^{n} w_{ni}(x) = \frac{1}{s_0 s_2 - s_1^2}\Big[s_2 \sum_{i=1}^{n} K\Big(\tfrac{X_i - x}{h}\Big) - s_1 \sum_{i=1}^{n} (X_i - x) K\Big(\tfrac{X_i - x}{h}\Big)\Big].$$

But

$$\sum_{i=1}^{n} K\Big(\tfrac{X_i - x}{h}\Big) = s_0, \qquad \sum_{i=1}^{n} (X_i - x) K\Big(\tfrac{X_i - x}{h}\Big) = s_1,$$

so

$$\sum_{i=1}^{n} w_{ni}(x) = \frac{1}{s_0 s_2 - s_1^2}(s_2 s_0 - s_1 s_1) = 1.$$

Therefore, the weights always sum to one.

Practical exercise

Before turning to the simulation study, let us recall the connection with the theoretical part. The AMISE-optimal bandwidth h_{AMISE} depends on two unknown quantities, the variance σ^2 and the curvature term θ_{22}.
In practice, we estimate them using blockwise polynomial fits, and we select the number of blocks N via Mallow's C_p criterion.
The goal of the practical part is to study how h_{AMISE} behaves when changing the key parameters: the sample size n, the block number N, and the distributional parameters (α, β) of the Beta covariates.
All results are obtained from the interactive Shiny App (`app.R`), which implements the core functions coded in `functions.R` and provides an immediate visualization of how the estimated quantities and h_{AMISE} evolve when changing the parameters.
Below we discuss the main plots and observations.

Summary Table

The summary tab of the App reports the used number of blocks (either set manually or selected via the C_p criterion), together with the estimates $\hat{\sigma}^2$ and $\hat{\theta}_{22}$, and the corresponding h_{AMISE}. From these results, we observe that increasing N tends to increase $\hat{\theta}_{22}$, which in turn reduces h_{AMISE}.
This is expected because with very few blocks, the piecewise polynomials oversmooth and underestimate curvature, leading to a larger bandwidth. As N grows, curvature is captured more accurately and the estimated optimal bandwidth becomes smaller.
The C_p criterion is designed to balance this trade-off between bias (too few blocks) and variance (too many blocks).

Bandwidth vs Number of Blocks

The plot of h_{AMISE} against the number of blocks N shows a clear monotone decrease: as N increases from 1 to 5, the estimated bandwidth becomes smaller.

This behavior is explained by the role of $\hat{\theta}_{22}$ in the formula

$$h_{AMISE} = n^{-1/5} \left(\frac{35\, \hat{\sigma}^2\, |supp(X)|}{\hat{\theta}_{22}} \right)^{1/5}.$$

With very few blocks, the piecewise polynomials cover large regions of the support and tend to oversmooth, which underestimates the curvature $\hat{\theta}_{22}$.
Since h_{AMISE} is inversely related to $\hat{\theta}_{22}$, this results in an inflated bandwidth.
As N increases, the fits become more local and capture more curvature, so $\hat{\theta}_{22}$ increases and h_{AMISE} decreases accordingly.

In theory, if N were pushed much higher or n were smaller, one could expect unstable estimates of $\hat{\sigma}^2$ and $\hat{\theta}_{22}$ due to very few observations per block.
However, in our App N is restricted to at most 5 and the sample sizes are relatively large, so each block remains well populated and the estimates are stable.
As a result, the dominant effect observed in the App is the monotone decrease of h_{AMISE} with N.

Bandwidth vs Sample Size

The App also allows us to vary the sample size n and visualize the resulting h_{AMISE}.
We observe that the bandwidth decreases as n increases, in line with the theoretical rate

$$h_{AMISE} \propto n^{-1/5}.$$

This illustrates the classical bias–variance trade-off: with more data, we can afford to use a narrower bandwidth, thereby reducing bias while keeping variance under control.
It also explains why the optimal number of blocks N naturally depends on n: larger samples allow a finer partitioning of the support while still maintaining enough observations per block for stable estimation.

Effect of the Beta Distribution

Two additional plots in the App focus on the role of the distribution of $X \sim \text{Beta}(\alpha, \beta)$.
The first shows the empirical distribution of X, while the second displays h_{AMISE} across different (α, β) values.

The shape of the Beta distribution clearly influences the estimated bandwidth.
When α or β is small, the distribution becomes highly skewed and most of the observations concentrate near the boundaries of the support, leaving other regions with relatively few points.
In theory, such imbalanced designs can create problems: blocks that contain very few observations provide unstable polynomial fits, which in turn affect the estimates of $\hat{\sigma}^2$ and $\hat{\theta}_{22}$.

In practice, however, our implementation always fits a polynomial as long as there are at least six points in a block.
As a result, the App never shows empty or grey regions: the fits are produced everywhere, even in low-density areas.
This makes the visual output smooth and continuous, but it also hides the fact that estimates in these regions may be fragile, since they rely on very limited information.

Overall, the plots confirm that the covariate distribution has a strong impact on h_{AMISE}.
Highly skewed Beta distributions amplify the risk of poorly estimated curvature and variance, even though the App construction ensures that a fit is always returned.

Data and Fit

The last plot in the App compares the simulated data with the true regression function $m(x)$ (blue line) and the blockwise polynomial fit (red line).

When the number of blocks N is small, the fitted polynomial curves oversmooth and fail to capture the oscillatory shape of the true regression function. Increasing N improves the fit, as the polynomials become more local and can adapt better to the curvature of $m(x)$.
However, the quality of the fit also depends on the number of observations within each block. With the relatively large sample sizes considered in the App and the cap of $N \leq 5$, each block still contains enough data for stable estimation, so we do not observe extreme instability.

This visualization provides a clear confirmation of the bias–variance trade-off:
- too few blocks lead to biased, overly smooth fits,
- too many blocks could lead to unstable variance estimates if n were smaller.

In our setting, the App illustrates how the C_p criterion automatically selects a reasonable value of N that balances these two effects.

Conclusions

Through the Shiny App exploration, we investigated how the AMISE-optimal bandwidth h_{AMISE} depends on key factors:
the block number N, the sample size n, and the distribution of the covariates (α, β).

The main findings are as follows:

- **Effect of N**: Increasing the number of blocks reduces h_{AMISE} in our simulations, as more curvature is captured through $\hat{\theta}_{22}$. For small N, the estimator oversmooths and overestimates the bandwidth, while larger N values provide more flexibility. In practice, C_p provides a principled compromise between bias and variance.
- **Effect of n**: The bandwidth decreases with larger sample sizes, consistent with the theoretical rate $h_{AMISE} \propto n^{-1/5}$. With more observations, narrower bandwidths can be used while controlling variance.
- **Effect of the Beta distribution:** The parameters (α, β) strongly influence the density of the covariates. Highly skewed Beta distributions produce regions with few observations, where estimates are less reliable. In our implementation, however, a fit is always produced as long as minimal data are available, so these weaknesses are not directly visible in the App.
- **Data and fit:** Visual comparison between the true function and the blockwise polynomial fit highlights the bias–variance trade-off. Too few blocks lead to overly smooth fits that miss local features, while too many blocks could in principle cause instability. The C_p criterion balances these effects and selects a reasonable value of N.

Overall, the Shiny App provides an intuitive and interactive way to understand the trade-offs in bandwidth selection.
It confirms the theoretical predictions while also illustrating practical issues related to the number of blocks, sample size, and covariate distribution.

All these results can be reproduced and explored interactively in the Shiny App (`app.R`).
The code of the App as well as the functions used for the simulations (`functions.R`) are included below to ensure full reproducibility.